

- 5 Phenotypic variation exists in natural populations. There are many causes of variation. Natural selection determines which phenotypes are advantageous.

(a) Variation in a particular characteristic can be described as either discontinuous or continuous.

Table 5.1 contains a list of statements that apply to discontinuous variation, continuous variation or both.

Complete both columns of Table 5.1. Put a tick (✓) in the box if the statement applies and leave the box empty if the statement does not apply.

Table 5.1

statement	discontinuous variation	continuous variation
often involves one gene only		
environmental factors may affect gene expression		
there is an additive effect of genes that contributes to the phenotype		
there are distinct differences between the various forms of a characteristic		

[4]

- (b) There is variation in the quantity of vitamin D stored in the body.

Vitamin D has an important role in keeping bones healthy. The main storage form of vitamin D in the body is serum 25-hydroxyvitamin D (serum 25-OHD).

A study was carried out on 262 healthy women to investigate if the concentration of serum 25-OHD varied between summer and winter. The women had taken no vitamin D supplements. The age range of the women in the sample was 40 to 72 years old.

Table 5.2 shows the results of the study.

Table 5.2

group	mean concentration of serum 25-OHD /ng cm ⁻³	standard deviation
sampled during summer $n = 138$	32.7	7.6
sampled during winter $n = 124$	28.5	8.3
whole sample $n = 262$	30.7	8.2

Additional analysis showed that there was no significant correlation between age and serum 25-OHD concentration.

- (i) Explain what is meant by standard deviation.

.....
 [1]

- (ii) The t -test was used to compare the mean concentration of serum 25-OHD when sampled during the summer with the mean concentration of serum 25-OHD when sampled during the winter, as shown in Table 5.2.

Calculate the value of t using the formula provided.

$$t = \frac{|\bar{x}_1 - \bar{x}_2|}{\sqrt{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)}}$$

key to symbols:

$\bar{x}$ = mean

s = sample standard deviation

n = sample size (number of observations)

Give your answer to **four** significant figures.

There is space for your working.

t -test value [3]

[illegible]

[3]

..... [1]

1

1